



Discovery of novel virulence mechanisms in *Clostridium botulinum* type A3 using genome-wide analysis

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ABSTRACT

Objective: *Clostridium botulinum* type A is a neurotoxin-producing, spore-forming anaerobic bacterium that causes botulism in humans. The evolutionary genomic context of this organism is not yet known to understand its molecular virulence mechanisms in the human intestinal tract. Hence, this study aimed to investigate the mechanisms underlying virulence and pathogenesis by comparing the genomic contexts across species, serotypes, and subtypes.

Methods: A comparative genomic approach was used to analyze evolutionary genomic relationships, inter-genomic distances, syntenic blocks, replication origins, and gene abundance with phylogenomic neighbors.

Results: Type A strains have shown genomic proximity to group I strains with distinct accessory genes and vary even within subtypes. Phylogenomic data showed that type C and D strains were distantly related to a group I and group II strains. Synthetic plots indicated that orthologous genes might have evolved from *Clostridial* ancestry to subtype A3 strains, whereas syntonic out-paralogs might have emerged between subtypes A3 and A1 through α -events. Gene abundance analysis revealed the key roles of genes involved in biofilm formation, cell-cell communication, human diseases, and drug resistance compared to the pathogenic *Clostridia*. Moreover, we identified 43 unique genes in the type A3 genome, of which 29 were involved in the pathophysiological processes and other genes contributed to amino acid metabolism. The *C. botulinum* type A3 genome contains 14 new virulence proteins that can provide the ability to confer antibiotic resistance, virulence exertion and adherence to host cells, the host immune system, and mobility of extrachromosomal genetic elements.

Conclusion: The results of our study provide insight into the understanding of new virulence mechanisms to discover new therapeutics for the treatment of human diseases caused by type A3 strains.

1. Introduction

Clostridium botulinum, a spore-forming, Gram-positive anaerobic bacterium, is not the only *Clostridia* species capable of producing botulinum toxins (BoNTs) but also *C. butyricum* and *C. baratii* are rarely associated with botulism intoxication. Eight antigenic variants (A, B, C1, D, E, F, G, and H) of BoNTs have been recognized. These serotypes can be divided into at least forty subtypes. BoNTs cause flaccid paralysis of peripheral nerve terminals, leading to lethal conditions known as botulism (Rossetto et al., 2014). BoNTs produced by vegetative cells

block acetylcholine release in intoxicated hosts and do not react with any substrate in the presynaptic motor neurons. *C. botulinum* spores have been found in soil, honey, aquatic sediments, plants, and clinical samples that are linked to human botulism (Williamson et al., 2016). Spores are resistant to desiccation, heat, ultraviolet light, and alcohol and can survive boiling for up to 4 h. The four types of naturally occurring human botulism include food-borne, infant, wound, and inhalation botulism. Homemade canned, preserved, or fermented foodstuffs are common sources of food-borne botulism. Inhalation botulism has a clinical footprint that is similar to that of food-borne botulism.

Abbreviations: BoNTs, Botulinum toxins; Ka/Ks, Synonymous and non-synonymous substitutions; CPE, *C. perfringens* 13 (BA000016); CPR, *C. perfringens* SM101 (CP000312); CTC, *C. tetani* E88 (AE015927); CTET, *C. tetani* 12124569 (HG530135); CNO, *C. novyi* NT (CP000382); CBV, *C. botulinum* A2 Kyoto (CP001581); CLD, *C. sporogenes* NCIMB 10696 (CP009225); CBUT, *C. butyricum* KNU-L09 (CP013252); CPF, *C. perfringens* ATCC 13124 (CP000246); CDL, *C. difficile* R20291 (FN545816); PDF, *C. difficile* 630Derm (LN614756); CST, *Acetanaerobium sticklandii* DSM 519 (FP565809); EAC, *Peptoclostridium acidaminophilum* DSM 3953 (CP007452); HPY, *Helicobacter pylori* 26695 (AE000511); HHE, *H. hepaticus* ATCC 51449 (AE017125).

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According to the Centers for Disease Control and Prevention, approximately 37,783 cases of food-borne illnesses have been reported between 2018 and 2020, of which 2226 are reported annually for human botulism (<https://www.cdc.gov/norsdashboard>).

C. botulinum classification into four groups is not based on their BoNTs but on their metabolic and phenotypic features. The Group I strains contain serotypes A, B, and F, whereas group II strains contain serotypes B, E, and F, which cause botulism in humans. Group III strains harbor C and D serotypes that cause botulism in birds and animals. Type-G is included in the Group IV strains responsible for human botulism. Type A strains are of great concern worldwide because they cause food-borne, infantile, and wound botulism (Chellapandi and Prisilla, 2018). Type A strains are accountable for 963 annual botulism cases worldwide, of which 42% are caused by infant botulism and 79% by wound botulism (CDC, 2013). Foodborne illnesses affect more than 76 million people in the United States annually. Death is caused by doses of the oral route at 30 ng, by the inhalational route at 0.80 to 0.90 µg, and by the intravenous route at 0.09 to 0.15 µg (Sobel, 2005; Peck, 2006). Antibiotics have no activity against the toxins themselves. Hence, botulinum toxin is classified as a class I bio-weapon that includes deliberate contamination of food or beverages or aerosol release (Armonet et al., 2001; Villar et al., 2006; Dhaked et al., 2010).

Comparative genomic analyses have provided knowledge to understand the global distribution of strains expressing various toxin types. The BoNT gene cluster may have evolved from an ancestral collagenase-like gene cluster. Type A genomes are phylogenetically related to the non-foodborne bacterium *Clostridium sporogenes* (Carter et al., 2009, 2010). The pathogenicity, metabolic diversity, and evolution of this species are highlighted with comparative genomic analysis. It also revealed intergenomic evolutionary restrictions on metabolic activities and adjustments across *C. botulinum* genomes (Fang et al., 2011). Italian group I strains are heterogeneous and are distributed in different lineages owing to their high recombination rates (Giordani et al., 2015). Type A genomes harbor five botulinum toxin subtypes A1–A5 (BoNT/A), of which Hall A Sanger is a well-known strain studied in detail. Genome size varies within subtype A1 strains, and 90% of the core genes are shared across them (Ng and Lin, 2014).

Microbial pathogenesis is mediated by secretion machines, virulence regulatory networks, and secreted virulence effectors (Beiting and Roos, 2011; Yoon et al., 2011). Clostripain (*cloSI*), α-hemolysin (*hlyA*), collagenase (*colA*), and adhesion proteins are virulence factors that facilitate the ability of *C. botulinum* type A to establish itself on or within a host (Chen et al., 2016). The virulence factors produced by this organism can determine the degree of molecular virulence and pathogen-host interactions in disease states. *C. botulinum* type Hall type A1 strain contains six new virulence factors underlying its pathogenesis in the host (Chellapandi and Prisilla, 2018), and there are no reports on other subtypes. Gene homology analysis revealed unique clades of type A strains formed from several toxin subtypes (A1–A5) and some type A strains also harbor unexpressed genes (Raphael et al. 2014). The chromosome is a neurotoxin cluster location for the A1 gene of ATCC3502, A2 gene of Kyoto-F, and A5 gene of H04402 065. The neurotoxin cluster locus of these subtypes is *oppA/brnQ*. Conversely, plasmid and plasmid-borne sites are neurotoxin cluster locations and locus for the A3 gene of Loch Maree, respectively (Marshall et al., 2007). The DNA sequence identity of the different subtypes can also vary from 2 to 333 nucleotides (Carter and Peck, 2015). Compared with the prototype BoNT/A1, subtype A3 is the most divergent subtype of BoNT/A with a 15.4% difference at the amino acid level (Mad'arova et al. 2017). *C. botulinum* subtype A3 caused eight deaths from glass-potted duck paste at Loch Maree hotel in the United Kingdom (Horowitz, 2011). This subtype has also been associated with food-borne botulism in Slovakia, 2015 (Mad'arova et al., 2017).

Hence, the present study aimed to develop a phylogenomic framework based on knowledge of molecular diversity, evolutionary relationships, and genomic contexts across serotypes and subtypes of these

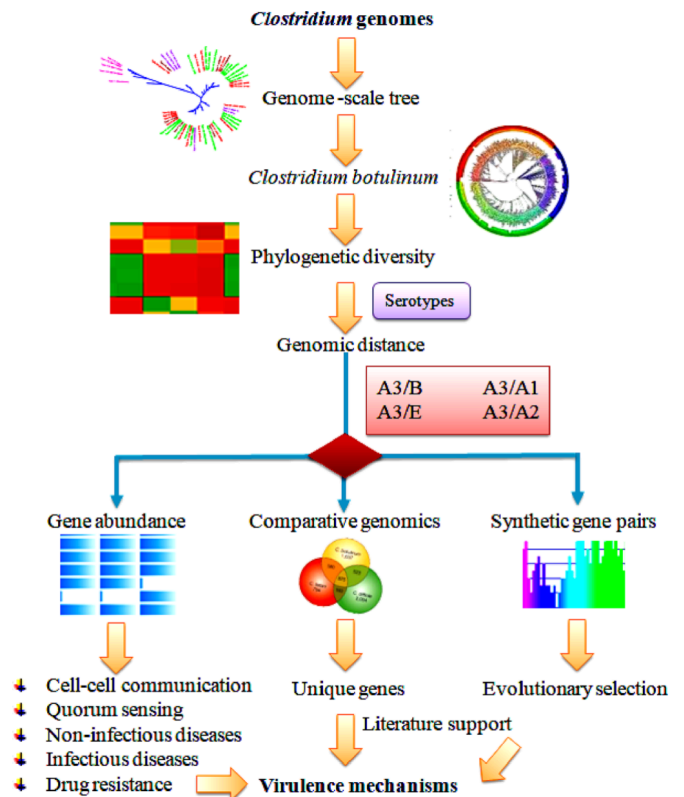


Fig. 1. Flow chart of study to predict the virulence mechanisms of *C. botulinum* type A.

genomes. This study also characterized differences among *C. botulinum* genomes and compared differentiating genomic features with genome-wide analyses. The present study predicted gene abundance in cell–cell communication, human diseases, and drug resistance. This study analyzed genes on a case-by-case basis to improve genome annotation and to identify unique genes, enzymes, and new virulence proteins. The results of our study reveal their phylogenetic proximity and metabolic discrepancy for new drugs and target discovery against human botulism.

2. Materials and methods

2.1. Dataset

Whole genome sequences of *C. perfringens* 13 (BA000016), *C. perfringens* SM101 (CP000312), *C. tetani* E88 (AE015927), *C. tetani* 12124569 (HG530135), *C. novyi* NT (CP000382), *C. botulinum* A2 Kyoto (CP001581), *C. sporogenes* NCIMB 10696 (CP009225), *C. butyricum* KNU-L09 (CP013252), *C. perfringens* ATCC 13124 (CP000246), *C. difficile* R20291 (FN545816), *C. difficile* 630Derm (LN614756), *Acetanaerobium sticklandii* DSM 519 (FP565809), *Peptoclostridium acidaminophilum* DSM 3953 (CP007452), *Helicobacter pylori* 26695 (AE000511) and *H. hepaticus* ATCC 51449 (AE017125) were retrieved from the National Collection for Biotechnology Information (NCBI) (<https://www.ncbi.nlm.nih.gov>). The complete genome sequences of *C. botulinum* A ATCC 3502 (AM412317), *C. botulinum* A2 Kyoto (CP001581), *C. botulinum* A3 Loch Maree (CP000962), *C. botulinum* B Eklund 17B (CP001056) and *C. botulinum* F 230613 (CP002011) were used for comparative genomic analysis within species. We used genome annotation data of these organisms available in public domain databases for functional categorization of protein-coding genes. The comparative genomic approach used for the prediction of new virulence mechanisms of *C. botulinum* type A is depicted in Fig. 1.

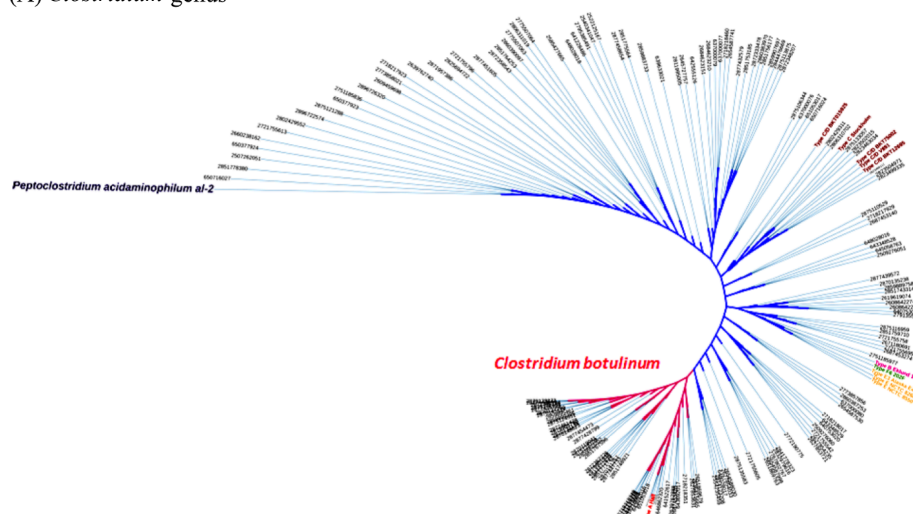
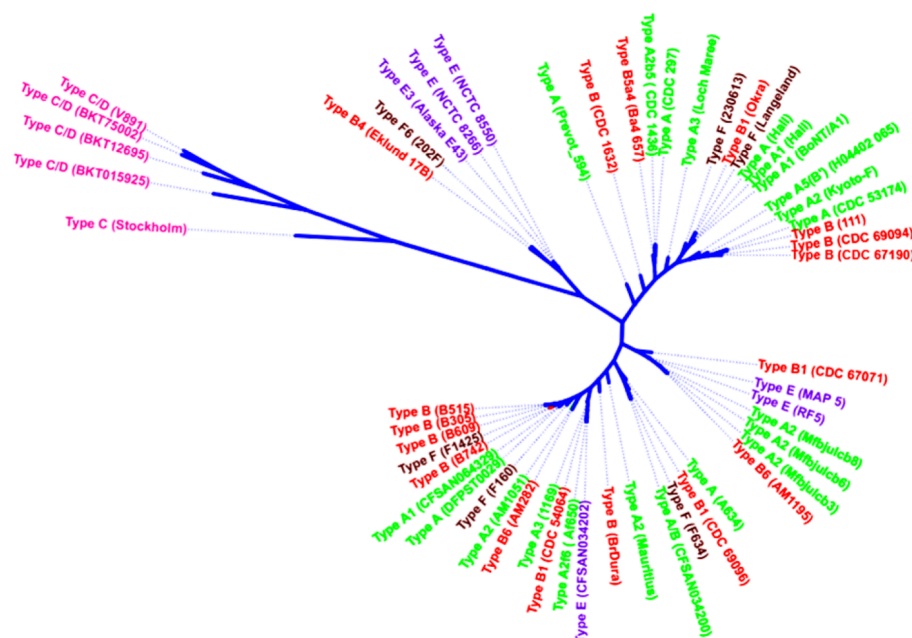
(A) *Clostridium* genus(B) *Clostridium botulinum*

Fig. 2. Phylogenomic tree constructed using a compositional vector algorithm from the complete genome sequences of *Clostridium* sp. Branch lengths are proportional to evolutionary distances. The correlation matrix represents (Fig. 2C) the intergenomic distances calculated between *C. botulinum* type A3 and other strains.

2.2. Phylogenomic analysis

The phylogenomic trees were constructed from hidden evolutionary information in the genomic context of 208 *Clostridium* species and 57 *C. botulinum* to determine their evolutionary relationships using the CVTree v4.0 server (Zuo, 2021). The datasets included only finished genome sequences for phylogenomic analysis of this study (Supplementary file). We excluded draft, unfinished, and microbiome genome sequences in the datasets. The alignment-free algorithm was used to infer phylogenetic relationships from the genome sequences. The statistical re-sampling tests such as Jack-knife and bootstrap tests were neglected. The K-tuple length was set as 6 with modified lineage. Average genome size and gene counts were 38, 45, 899 bp, and 3, 903 bp, respectively. The subtraction procedure based on K-tuple length order was used to overpower the consequence of neutral mutations and to highpoint the decisive role of natural selection. The neighbor-joining algorithm was used to calculate distances for all species pairs comprising

a dissimilarity matrix to construct phylogenomic trees (Saitou and Nei, 1987). The constructed tree was rooted with *Peptoclostridium acidaminophilum* al-2 to determine which in-group was most likely to have a specific evolutionary change (Galperin et al., 2016). The final tree was visualized using the iTOL v3.0 viewer (Letunic and Bork, 2016). The genomic correlation coefficient was calculated to determine phylogenomic neighbors across *Clostridium* genomes using the Integrated Microbial Genomes and Microbiomes v5.0 (IMG-M) genome comparison tool (Chen et al., 2019). It uses a dissimilarity matrix, which is composed of collections of distances between genome pairs. The intergenomic distances between the genomes were calculated based on the Pearson correlation coefficients (Benesty et al., 2009). The functional profiles of selected genomes were analyzed based on clusters of orthologous groups.

(C) Intergenomic distances

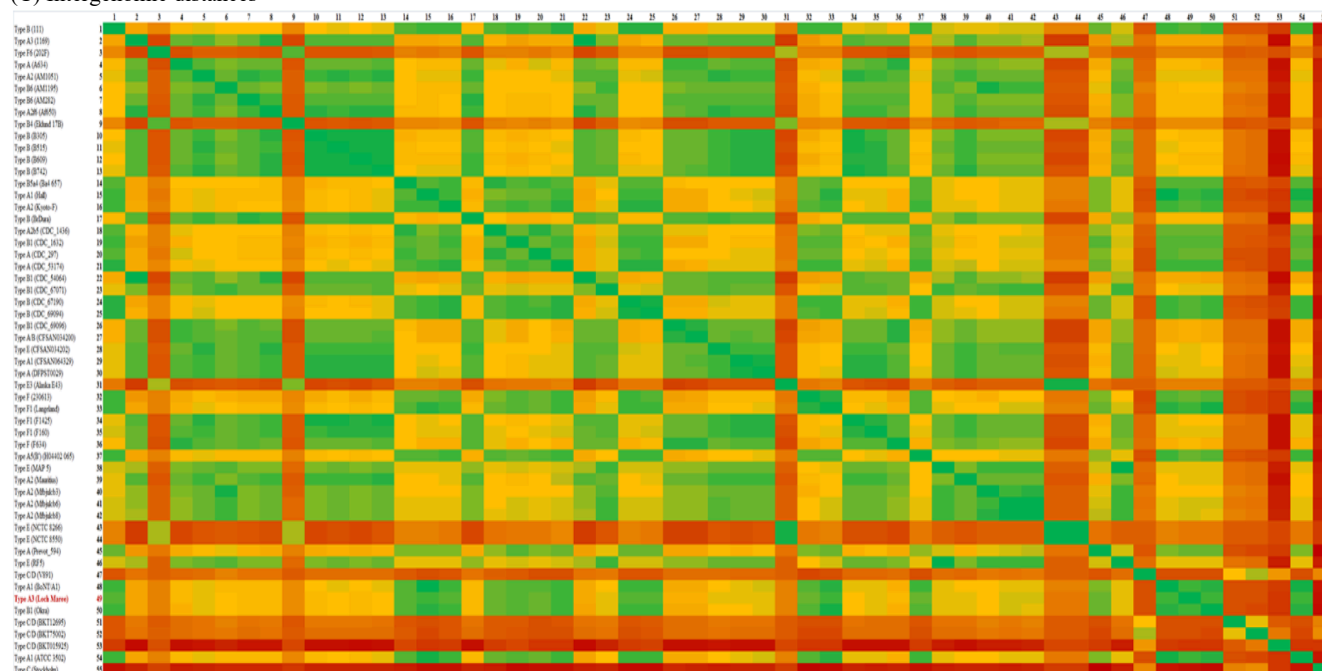


Fig. 2. (continued).

2.3. Analysis of genome syntenic pairs

Phylogenetic diversity, gene order, genome organization, and arrangements are modified by a typical evolutionary process. Syntenic plots were constructed to compare the protein-coding genes of the A3 genome across serotypes B and E; subtypes A1 and A2 using CoGe: SynMap (Haas et al., 2004). Nucleotide-nucleotide searching was performed with a LastZ algorithm (<https://last.cbrc.jp>). The relative gene order was used to calculate nucleotide distance where the maximum distance between two matches was set as 20 genes and a minimum number of aligned pairs set as 5 genes. Syntenic blocks were computed using the interactive DAGChainer algorithm (Haas et al., 2004). Syntenic depth was calculated using the Quota Align algorithm with a ratio of coverage depth ranging from 1 to -1 and an overlapping distance of 40 (Tang et al., 2011). For functional bias, the maximum number of chromosomes in the analysis was 25, and skipped random/unknown chromosomes. All genes in the target genome were analyzed for functional bias. Syntenic coding domain sequence pairs were identified using the CodeML algorithm (Gao et al., 2019). The tandem duplication distance was set as 10. Syntenic blocks and evolutionary events resulting from synonymous and non-synonymous substitutions (Ka/Ks) were represented as syntenic histograms. The genome rearrangements were analyzed by computing Ka/Ks values between syntenic gene pairs. The genomic compositional dissimilarity was measured at a non-metric multidimensional scale by the Bray-Curtis dissimilarity coefficient (Bray and Curtis, 1957).

2.4. Comparative metabolic analysis

An extensive bioinformatics analysis was used to examine whether subtype A3 shared common genes, enzymes, and metabolic subsystems with other types and subtypes (Bharathi and Chellapandi, 2017a, 2017b; Prathiviraj and Chellapandi, 2020; Sangavai and Chellapandi, 2021). We gathered comprehensive information to determine the abundance of genes involved in biofilm formation, cell-to-cell communication, drug resistance, and communicable and non-communicable diseases manually from the Kyoto Encyclopedia of Genes and Genomes (KEGG) v104.0 database searching. Unique genes, metabolic enzymes, and

virulence proteins of the subtype A3 genome were identified by comparing the collected gene sets. The functions of the unique proteins identified from the subtype A3 genome were annotated based on the e-value criterion of 10^{-5} using KEGG-BLAST searches against *C. botulinum* genomes.

2.5. Discovery of novel virulence proteins

The virulence property of unique proteins identified through comparative genomic analysis was predicted from their amino acid sequences by the MP3 v2.0 tool. An integrated SVM-HMM approach was used for virulence prediction at a set threshold of -0.2 (Gupta et al., 2014). The most virulent proteins were selected based on a cut-off score ranging from 1.277 to 2.839. The antigenicity (vaccine) properties of the selected proteins were predicted using VaxiJen v2.0 (Doytchinova and Flower, 2007) at a set threshold of 0.4. The transmembrane helices of these proteins were identified using DeepTMHMM v1.0.20 (<https://dtu.biolib.com/DeepTMHMM>). Signal peptide and protein sorting were predicted from the sequences using SignalP v6.0 (Teufel et al., 2022) and PSORTb v2.0 (Gardy et al., 2005), respectively.

3. Results

3.1. Phylogenomic analysis

Phylogenies of the whole-genome sequences of *Clostridium* species were used to assess the evolutionary relatedness and diversity of botulinum toxin-producing strains (Fig. 2A). This shows that the constructed tree consisted of 11 significant clusters comprising various *Clostridium* groups. It also showed evolutionary links between other subtypes of A genomes. *P. acidaminophilum* al-2 served as an outgroup organism for the taxonomic classification of *Clostridium* sp. The type A, B, and F genomes closely resemble each other. The type C and D genomes are phylogenetically divergent from other *C. botulinum* serotypes and might act as ancestors of other botulinum toxin-producing strains (Fig. 2B). The subtype A3 genome is closely related to subtype A1 (strain CDC 287), and then to the type F genome. The intergenomic distances and genomic functional correlations of type A genomes were more closely related to

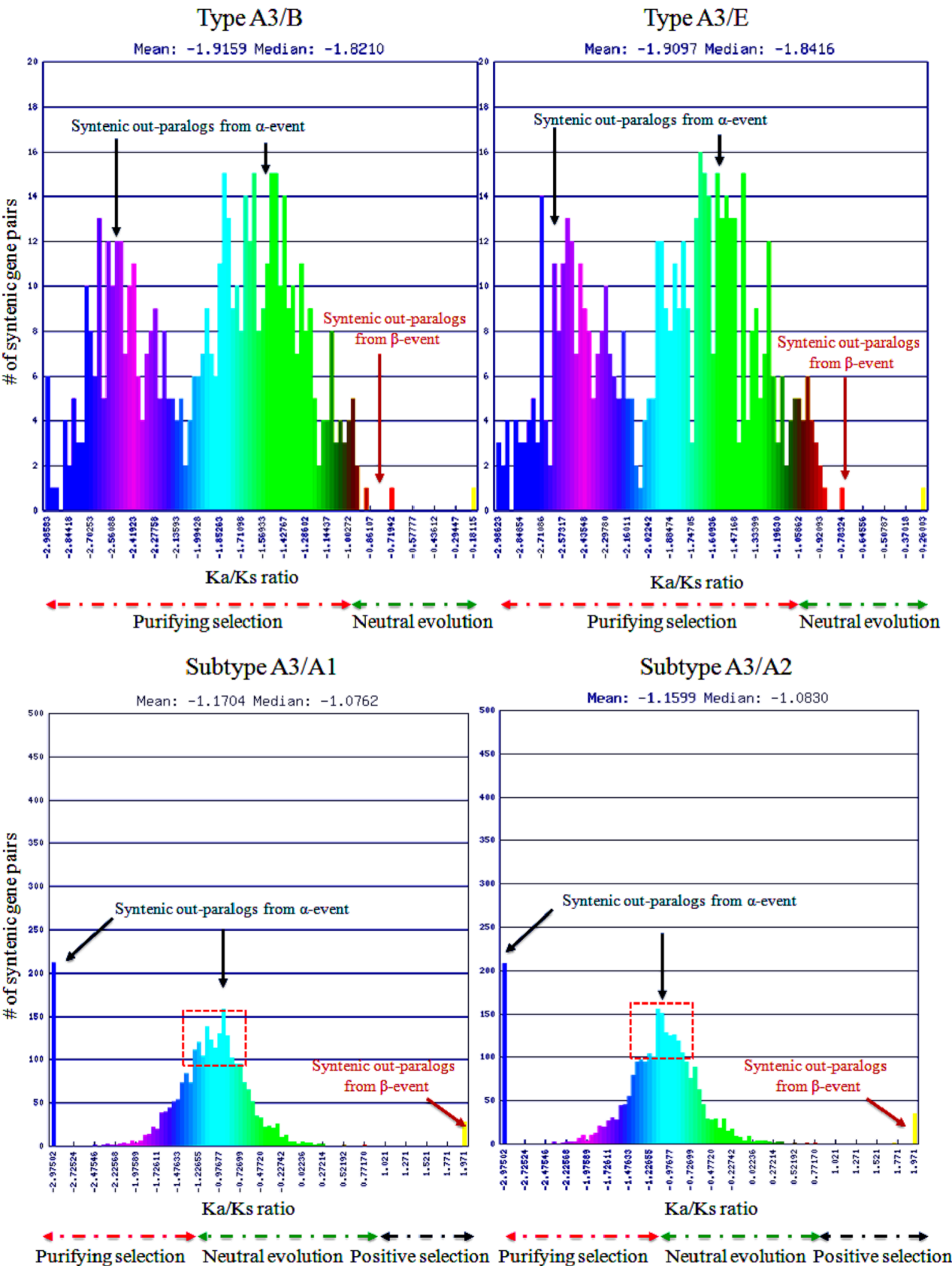


Fig. 3. Histogram of Ks/Ks ratio for evolutionary selection of syntenic gene pairs in subtype A3 from types and subtypes of *C. botulinum*.

Table 1

Genome-wide prediction of gene abundance in cell–cell communication, human diseases, and drug resistance from the pathogenic *Clostridia* of the human intestinal tracts.

Code	Genome	Genome ID	Cell-cell commutation	Non-infectious diseases	Infectious diseases	Drug resistance
CBH	<i>Clostridium botulinum</i> A Hall	NC_009495	70	18	7	14
CPE	<i>Clostridium perfringens</i> 13	NC_000915	45	19	9	15
CPR	<i>Clostridium perfringens</i> SM101	NZ_CP075979	44	19	6	13
CTC	<i>Clostridium tetani</i> E88	NC_004557	63	21	9	13
CTET	<i>Clostridium tetani</i> 12124569	NC_022777	62	20	9	11
CNO	<i>Clostridium novyi</i> NT	NZ_CP029458	81	16	8	16
CBV	<i>Clostridium baratii</i> Sullivan (serotype F7)	NZ_CP014204	50	20	13	24
CLD	<i>Clostridium sporogenes</i> NCIMB 10696	NZ_CP084662	20	6	1	9
CBUT	<i>Clostridium butyricum</i> KNU-L09	NZ_CP013239	92	19	15	18
CDC	<i>Clostridioides difficile</i> CD196	NZ_CP076401	46	21	6	15
CDL	<i>Clostridioides difficile</i> R20291	NZ_CP076401	79	20	14	22
PDF	<i>Clostridioides difficile</i> 630Derm	NZ_CP076401	92	19	14	22
CST	<i>Acetanaerobium sticklandii</i> DSM 519	NC_014614	84	20	1	15
EAC	<i>Peptoclostridium acidaminophilum</i> DSM 3953	NZ_CP007452	67	15	9	15
HPY	<i>Helicobacter pylori</i> 26695	NZ_CP071982	42	13	32	12
HHE	<i>Helicobacter hepaticus</i> ATCC 51449	NC_004917	44	12	9	11

Note: HPY and HHE were used as the reference genomes for comparative analysis.

those of type B genomes, followed by those of types F and E genomes (Fig. 2C). Type A, B, F, and E genomes share a close evolutionary relationship despite minor metabolic differences. Additionally, the subtype A3 and A1 genomes have similar genetic relationships, while the type C and D genomes have varied genetic relationships.

3.2. Analysis of genome syntenic pairs

Using genome syntenic analysis, we discovered gene features and collinearity shared between the serotypes and subtypes (Fig. 3). This indicated that the majority of genes in the A3 genome replicated at a high rate of synonymous and non-synonymous changes. Syntonic blocks modify the collinearity of either gene orders or gene families at the sites of these syntonic out-paralogs. The collinearity collection of genes shared between subtype A3 and type E was greater than that shared between subtype A3 and type B. The rate of evolution was low, and core genes may not have evolved within the A3, B, and E genomes. Among the A1, A2, and A3 subtypes, accessory genes may have evolved to establish in the host cell environment by positive selection through β -events. Syntonic out-paralogs are more likely to emerge between subtypes A3 and A1 through α -events than between subtypes A3 and A2 through β events. Orthologous genes in the A3 genome may have evolved from ancestral genomes and were subsequently established to retain unique growth physiology, colonization, and virulence in the human intestinal tract.

3.3. Comparative metabolic analysis

Gene abundance analysis indicated the pathophysiological roles of type A strains in biofilm formation, cell–cell communication, human disease, and drug resistance (Table 1). Forty-six genes were associated with biofilm formation in CNO, CBUT, CDL, and CST, the majority of which were involved in pilin formation (Fig. 4). The most pathogenic *Clostridia* had higher gene abundance (24 genes) for the quorum sensing process, suggesting cell-to-cell communication in the host environment. This study identified 18 genes in *Clostridial* genomes, in which CTC, CDL, and CST have more genes involved in non-infectious diseases in humans. *Clostridial* genomes contain 7 genes involved in human infectious diseases. Genes associated with drug resistance were more abundant (14 genes) in all the pathogenic *Clostridia*. CLD is a non-pathogenic bacterium with fewer antibiotic-resistance genes. This analysis indicated that at least one of the genes responsible for drug/antibiotic resistance is missing in pathogenic *Clostridia*.

3.4. Discovery of novel virulence proteins

Compared to other serotypes and subtypes, genome-wide analyses identified genes and enzymes that were not covered in the genome annotation data in the A3 genome (Fig. 5). The A3 genome contained 313 unique genes and 126 enzymes compared with the type B and E genomes. Among the type B and E genomes, 1072 genes and 503 enzymes were commonly shared, representing their evolutionary resemblance with the A3 genome. The A3 genome includes 16 unique enzymes and 49 genes, of which 1363 genes and 634 enzymes are shared across the A2 and A3 genomes. This suggests that *C. botulinum* genomes have phylogenetic proximity with metabolic complexity, which provides A3 strains with the ability to withstand their growth and adaptation in the gut environment. We identified 14 unique enzymes from the A3 genome compared to the A1 and A2 genomes, of which catalase and micrococcal nuclease have been predicted as virulence proteins (Table 2). These genes play key roles in amino acid metabolism, nucleotide metabolism, sugar metabolism, pentose phosphate pathway, DNA repair and recombination, terpenoid backbone biosynthesis, β -lactam resistance, and transport in A3 strains.

3.5. Discovery of novel virulence proteins

The functional categorization of the unique genes identified from the A3 genome across the A1 and A2 genomes (Table 3). This shows that the A3 genome consists of 29 unique genes involved in cell-to-cell communication, host defense system, antibiotic resistance, and drug targets. Pilus assembly protein, serine/threonine protein phosphatase 1, histidine kinase, and tight adherence proteins B and C are cell-to-cell communication proteins that were predicted as virulence proteins of the A3 genome. Death on curing protein, plasmid segregation protein, and REP-associated tyrosine transposase are virulence proteins predicted by the host defense system. In antibiotic-resistance systems, bacteriocin exporters, epoxyqueuosine reductase, and membrane-associated proteins have been identified as virulence proteins.

4. Discussion

Molecular genetic approaches have been used to understand the evolutionary links between various *Clostridial* species (Dobritsa et al., 2018; Togo et al., 2018; Zhang et al., 2018). Microarray investigations, amplified fragment length polymorphism, pulsed-field gel electrophoresis, and multi-locus sequence typing have been used to determine the diversity of *C. botulinum* strains and identify clades of bacteria that share common genes (Williamson et al., 2016). With advancements in DNA

(A) Biofilm

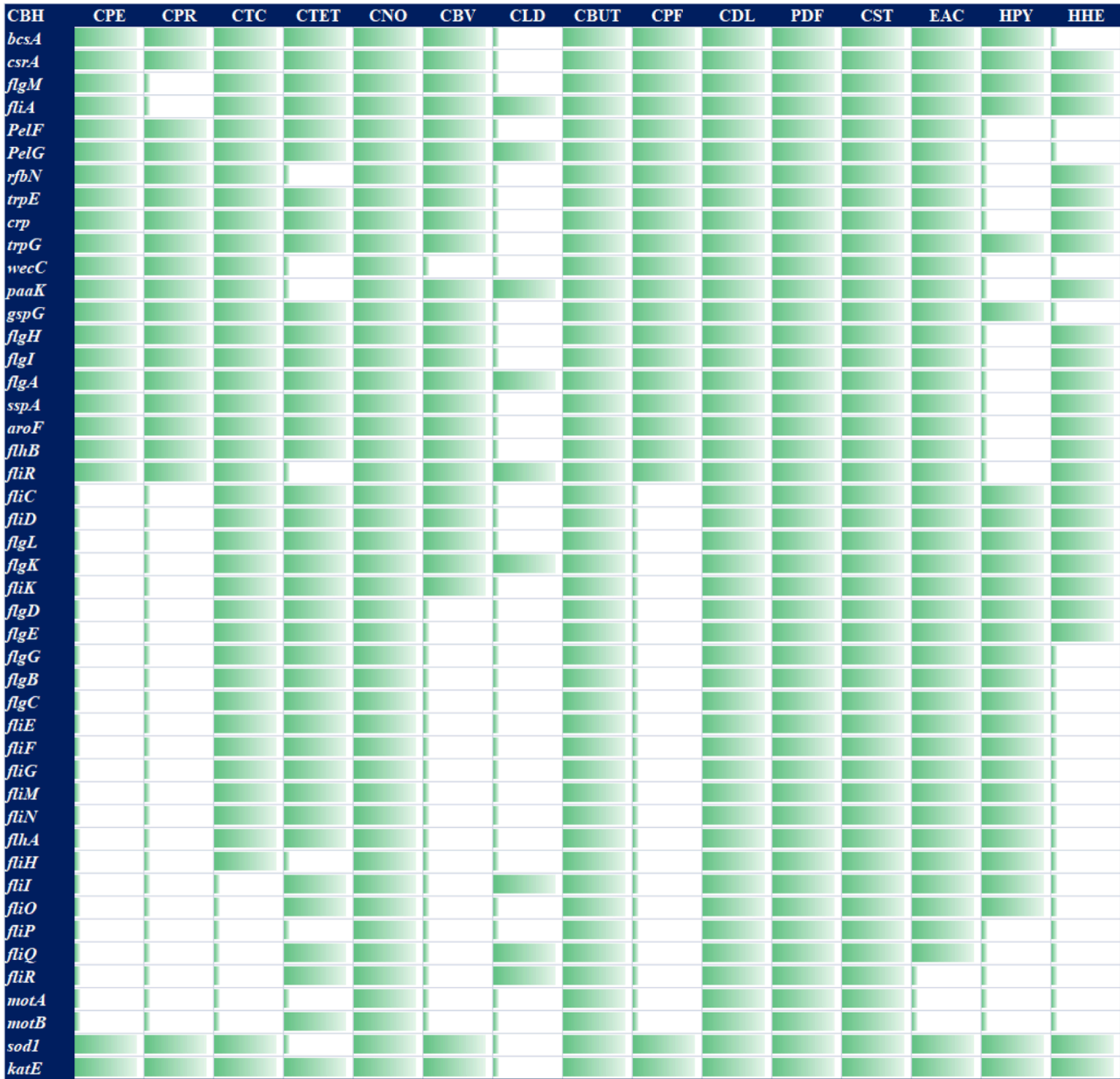
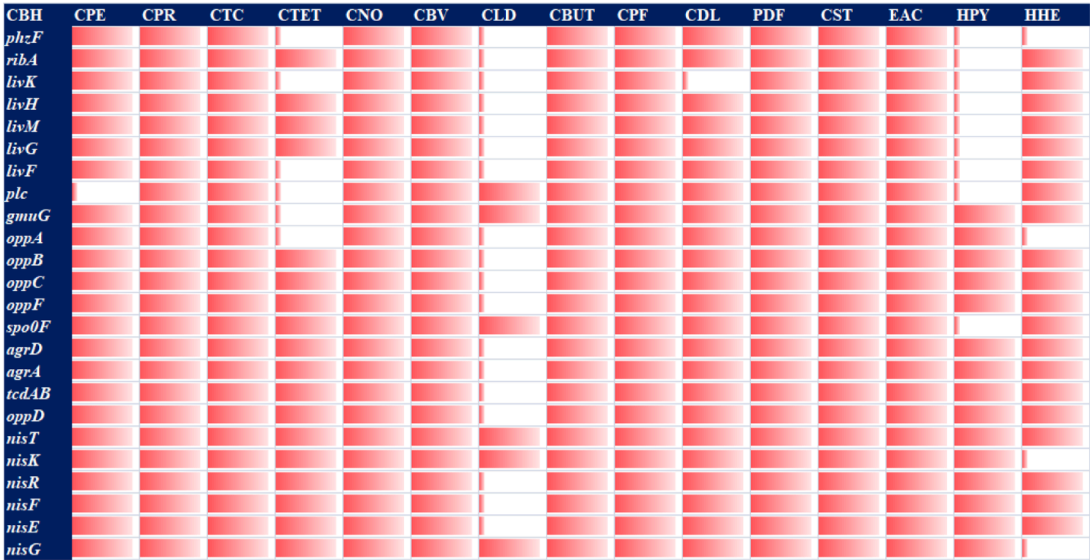


Fig. 4. Comparative genomic analysis of type A with pathogenic *Clostridia* for the prediction of its gene abundance in cell–cell communication, human diseases, and drug resistance. Abbreviation: CPE: *C. perfringens* 13 (BA000016), CPR: *C. perfringens* SM101 (CP000312), CTC: *C. tetani* E88 (AE015927), CTET: *C. tetani* 12,124,569 (HG530135), CNO: *C. novyi* NT (CP000382), CBV: *C. botulinum* A2 Kyoto (CP001581), CLD: *C. sporogenes* NCIMB 10696 (CP009225), CBUT: *C. butyricum* KNU-L09 (CP013252), CPF: *C. perfringens* ATCC 13124 (CP000246), CDL: *C. difficile* R20291 (FN545816), PDF: *C. difficile* 630Derm (LN614756), CST: *Acetanaerobium sticklandii* DSM 519 (FP565809), EAC: *Peptoclostridium acidaminophilum* DSM 3953 (CP007452), HPY: *Helicobacter pylori* 26,695 (AE000511), HHE: *H. hepaticus* ATCC 51449 (AE017125).

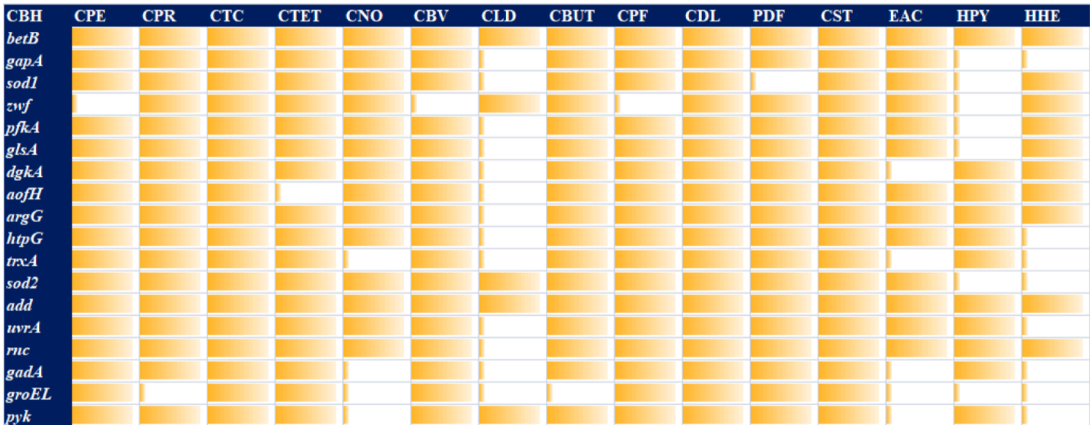
sequencing and bioinformatics tools, researchers can now demonstrate the evolutionary biology of prokaryotes and compare bacterial genomes at a higher resolution (Land et al., 2015; Bharathi and Chellapandi, 2017a, 2017b; Prathiviraj and Chellapandi, 2020; Sangavai and Chellapandi, 2021). The present study compared 208 whole-genome sequences of *Clostridium* sp., categorized their metabolic subsystems, and identified new virulence proteins. *C. botulinum* phylogenies require a solid classification system established by contrasting the phenotypic traits of the core genes. Therefore, proper positioning is a great challenge in describing the evolutionary relationships of *Clostridial* genomes (Schleifer, 2009; Varghese et al., 2015). The A3 genome was phylogenetically related to A1 and resembled the B and F genomes, but the type

C and D genomes were divergent from other serotypes. Type A, B, and F genomes belong to Group I, whereas type C and D genomes belong to Group III (Auch et al., 2010; Meier-Kolthoff et al., 2013). This implies that the BoNT gene cluster and accessory genes may have acted as evolutionary constraints on the *C. botulinum* phylogenetic lineage. Recombination, domain rearrangement, operon/gene cluster disruption, horizontal gene flux, and the gene order of a gene family determine the evolution of the bacterial genome using syntenic plots (Roberts and Smits, 2018; Castillo et al., 2018). Syntenic plots in our study indicated that core genes shared across a group I genomes showed ancestral characteristics and may not have evolved within subtypes, as the rate of evolution was low. Accessory genes may harbor virulence,

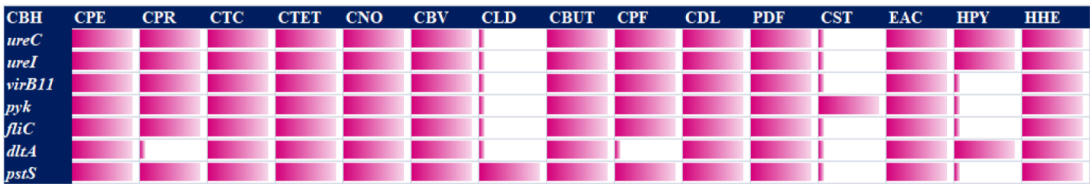
(B) Quorum sensing



(C) Non-infectious diseases



(D) Infectious diseases



(E) Drug resistance

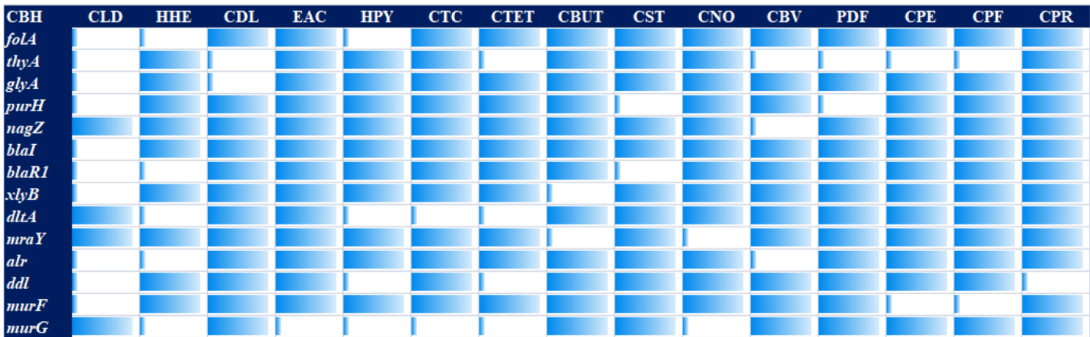


Fig. 4. (continued).

cell-to-cell communication, and host defense. These genes may have evolved in the A3 genome from A genome cluster through β -events, as described previously (Bharathi and Chellapandi, 2017a; 2017b;

Prathiviraj and Chellapandi, 2020; Sangavai and Chellapandi, 2021). Syntenic plot analysis revealed that A3 strains might acquire additional gene contexts to establish host-specific adaptation, colonization, and

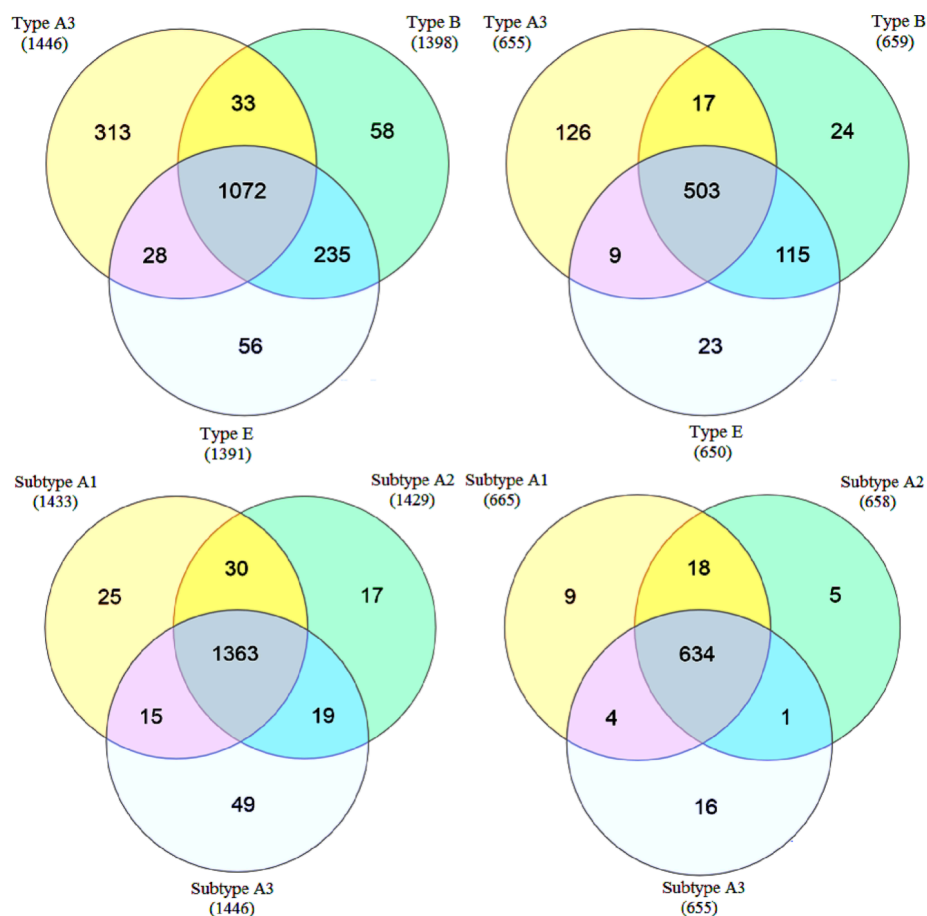


Fig. 5. Comparative genomic analysis of subtype A3 across other types and subtypes for finding its unique genes (Left) and enzymes (Right).

Table 2

Metabolic description of unique enzymes identified in the subtype A3 genome across the type A1 and A2 genomes.

Locus tag	Gene	EC	KO	Protein	Subsystem	Virulence status
CLK_2255	<i>katE</i>	1.11.1.6	K03781	Catalase	Tryptophan metabolism	P
CLK_1553	<i>ala</i>	1.4.1.1	K19244	Alanine dehydrogenase	Alanine, aspartate and glutamate metabolism	NP
CLK_2101	<i>legI</i>	2.5.1.101	K18430	N,N'-diacetyl legionaminase synthase	Amino sugar and nucleotide sugar metabolism	NP
CLK_2097	<i>per</i>	2.6.1.102	K13010	Perosaminesynthetase	O-Antigen nucleotide sugar biosynthesis	NP
CLK_1504	<i>phnW</i>	2.6.1.37	K03430	2-Aminoethylphosphonate-pyruvate transaminase	Amino acid-related enzymes	NP
CLK_0098	<i>ltrA</i>	2.7.7.49	K00986	RNA-directed DNA polymerase	Nucleotide metabolism	NP
CLK_2102	<i>legF</i>	2.7.7.82	K18431	CMP-N,N'-diacetyl legion aminic acid synthase	Amino sugar and nucleotide sugar metabolism	NP
CLK_0040	<i>pgl</i>	3.1.1.31	K07404	6-Phosphogluconolactonase	Pentose phosphate pathway	NP
CLK_0066	<i>nuc</i>	3.1.31.1	K01174	Micrococcal nuclease	Unclassified: metabolism	P
CLK_2100	<i>legG</i>	3.2.1.184	K18429	GDP/UDP-N, N'-diacetyl bacillosamine 2-epimerase (hydrolyzing)	Amino sugar and nucleotide sugar metabolism	NP
CLK_0062	<i>tag</i>	3.2.2.20	K01246	DNA-3-methyladenine glycosylase I	DNA repair and recombination proteins	NP
CLK_1596	<i>hyi</i>	5.3.1.22	K01816	Hydroxypyruvate isomerase	Glyoxylate and dicarboxylate metabolism	NP
CLK_0204	<i>idi</i>	5.3.3.2	K01823	Isopentenyl-diphosphate delta-isomerase	Terpenoid backbone biosynthesis	NP
CLK_0709	<i>rbsA</i>	7.5.2.7	K10441	Ribose transport system ATP-binding protein	Transporter	NP

pathogenesis in a new environmental niche.

Genome-wide comparative analysis has identified 43 unique genes in the A3 genome, of which 29 genes are involved in biofilm formation, the host defense system, antibiotic resistance, and drug targets, whereas 14 genes play key roles in diverse metabolic pathways, particularly amino acid metabolism. This study also predicted 14 new virulence proteins in the A3 genome that may exert higher virulence in addition to BoNT/A3 than other subtypes. The bacterial cell envelope is an essential barrier to environmental stress and contributes to virulence and antibiotic

resistance. A3 strains contain type IV pili assembly ATPase (*PilB*), which is similar to that of *C. difficile* and *C. perfringens* (Borriello et al., 1990; Varga et al., 2006; Proft and Baker, 2009; Shanmugasundarasamy et al., 2022). A balance between kinase and phosphatase activities of identified serine/threonine protein phosphatases may modulate the level of antibiotic resistance in A3 strains similar to *Enterococcus faecium* and *Staphylococcus aureus* (Sacco et al., 2014; Jarick et al., 2018). Histidine kinases in two-component bacterial systems are promising targets for novel antimicrobial and anti-biofilm agents. This protein has been

Table 3

Functional categorization of unique genes identified from the subtype A3 genome across the type A1 and A2 genomes.

Locus tag	Gene	KO	Protein function	Virulence status
Cell-to-cell communication				
CLK_0311	<i>cpaF</i>	K02283	Pilus assembly protein	P
CLK_0294	<i>virD4</i>	K03205	Type IV secretion system protein	NP
CLK_0238	<i>pphA</i>	K07313	Serine/threonine protein phosphatase 1	P
CLK_0309	<i>tadB</i>	K12510	Tight adherence protein B	P
CLK_0308	<i>tadC</i>	K12511	Tight adherence protein C	P
CLK_3435	<i>vanSC</i>	K18350	Histidine kinase-like protein	NP
CLK_2842	<i>arlS</i>	K18940	Histidine kinase	P
Host defense system				
CLK_0043	<i>doc</i>	K07341	Death on curing protein	P
CLK_0231	<i>pspA</i>	K03969	Phage shock protein A	NP
CLK_0227	<i>parM</i>	K18640	Plasmid segregation protein	P
CLK_1535	–	K07453	Putative restriction endonuclease	NP
CLK_0105	<i>rayT</i>	K07491	REP-associated tyrosine transposase	P
CLK_1522	<i>tnpB</i>	K07496	Putative transposase	NP
CLK_0143	<i>csH1</i>	K19114	CRISPR-associated protein 1	NP
CLK_0144	<i>csH2</i>	K19115	CRISPR-associated protein 2	NP
CLK_0145	<i>cas5h</i>	K19116	CRISPR-associated protein 5	NP
Antibiotic resistance mechanism				
CLK_2841	<i>dedA</i>	K03975	Membrane-associated protein	NP
CLK_2186	<i>pdxT</i>	K08681	Pyridoxal 5'-phosphate synthase	NP
CLK_0764	–	K09729	Membrane spanning protein	NP
CLK_2202	<i>queH</i>	K09765	Epoxyqueuosinereductase	P
CLK_3682	<i>blpA</i>	K20344	Bacteriocin exporter A	P
CLK_3683	<i>blpB</i>	K20345	Bacteriocin exporter B	P
CLK_0592	<i>tcaA</i>	K21463	Membrane-associated protein	P
CLK_1132	<i>yfiC</i>	K22900	tRNA (adenine37-N6)-methyltransferase	NP
CLK_3430	<i>vat</i>	K18234	Virginiamycin A acetyltransferase	NP
CLK_0017	<i>pbpD</i>	K18770	Penicillin-binding protein 4	P
Drug targets				
CLK_0712	<i>rbsB</i>	K10439	Ribose transport system substrate-binding protein	NP
CLK_0709	<i>rbsC</i>	K10440	Ribose transport system permease protein	NP
CLK_1552	<i>norG</i>	K18907	GntR family transcriptional regulator	NP

identified in A3 strains as an important factor regulating sporulation and enterotoxin production, as identified in *C. perfringens* (Bem et al., 2015; Freedman et al., 2019; King and Blackledge, 2021). Tight adherence proteins B and C of A3 strains are thought to play a role in host cell adhesion to damaged intestinal tissues, as evidenced by *C. perfringens* (Lepp et al., 2021). Therefore, the presence of surface proteins, two-component systems, and flagellum in type A3 strains is required for movement, adherence, and invasion of mucosal surfaces, as well as direct interaction with the host immune system (Vaishnavi, 2021).

Death on curing protein contains Fic domain with a conserved central motif (HPFx(D/E)GN(G/K)R). This protein phosphorylates Thr-382 of the translation elongation factor EF-Tu to inhibit bacterial translation (Castro-Roa et al., 2013; Cruz-García et al., 2014). It is considered a non-toxin virulence factor that contributes to disease establishment and host response in A3 strains, similar to multiple *C. difficile* virulence-associated pathways (Vedantam et al., 2012). Plasmids, bacteriophages, transposons, and pathogenicity islands of pathogenic bacteria harbor virulence factors that can be mobilized into different strains to create novel combinations of virulence factors (Desvaux et al., 2020). Plasmid segregation protein is part of a type II Par system in virulence plasmids. It controls the positioning of plasmid DNA during cell division so that plasmid-encoded toxin genes provide virulence flexibility when causing intestinal infections by A3 strains, as described for *C. perfringens* type A-E toxin plasmids (Freedman et al., 2015; Parreira et al., 2012). REP-associated tyrosine transposases identified from A3 strains are generally associated with high copy number sequences contributing to

host defense systems, such as enterobacteria (Nunvar et al., 2010).

Vaccination against human botulism in the general population is not warranted based on cost and expected adverse reactions. It also restricts therapeutic and cosmetic applications. The U.S. Food and Drug Administration did not license botulism toxoid vaccines for general distribution. Antitoxin therapy and some neurotoxin inhibitors (toosendanin) are more effective if administered early in the course of illness (Tacket et al., 1984). BoNT-based drug delivery vehicles can be used to directly deliver toosendanin to intoxicated nerve terminal cytosol to reverse paralysis (Zhang et al., 2009). A high level of resistance of *C. botulinum* has been observed against many antibiotics, including cycloserine, nitroimidazole, gentamycin, sulfamethoxazole, and trimethoprim. Cefoxitin and clindamycin showed relatively weak activities against *C. botulinum*. Antibiotics are not the preferred clinical treatment once symptoms of botulism arise (Chellapandi and Prisilla, 2018). Hence, the identification of new antibiotic-resistance genes, virulence factors, and drug targets can lead to the discovery of new antibiotics or inhibitors to prevent human botulism and associated diseases caused by A3 strains.

5. Conclusions

Phylogenomic proximity and metabolic discrepancies of the *C. botulinum* type A3 genome have been described intensively by comparing its genomic contexts within pathogenic *Clostridia*, serotypes, and subtypes. The phylogenomic data of this study showed good agreement with the current taxonomic classification of *Clostridial* genome lineages, as demonstrated by similar trends (Bharathi and Chellapandi, 2017a, 2017b; Prathiviraj and Chellapandi, 2020; Sangavai and Chellapandi, 2021). Neurotoxin cluster locus of the A3 gene of Loch Maree is a divergent subtype and located in a plasmid-borne site compared to other subtypes A. The subtype A3 strains have been associated with the recurrence of food-borne botulism in many countries recently. The results of our study revealed the phylogenomic equivalency of the subtype A3 genome with a group I genomes across serotypes, and with subtype A1 across subtypes. The subtype A3 genome shares a common core gene content and syntenic gene across serotypes and subtypes. The comparative genomic contexts analyzed its gene-phenotypic-metabolic characteristics and provided clues for the identification of new metabolic genes and virulence factors. The predicted surface, biofilm formation, two-component, pilin, plasmid, and transposon-carrying genes can afford subtype A3 strains for cell-to-cell communication, antibiotic resistance, host cell adherence, interaction with the host immune system, and virulence mobility from plasmids. To better understand these pathophysiological functions, we have to experimentally evaluate the precise functions of the predicted virulence factors using differential gene expression studies. This study will help us investigate new drugs or antibiotics against new virulence factors in the treatment of human botulism and associated diseases and to fill metabolic gaps in genome-scale metabolic reconstruction processes. However, its system-level characteristics will be useful for organism-specific therapeutic discovery and the development of personalized medicine for *C. botulinum* type A3-associated diseases.

Ethics approval and consent to participate:

The need for ethical approval and individual consent was waived.

Declaration of Competing Interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: [CHELLAPANDI P reports financial support was provided by Tamil Nadu State Council for Higher Education. CHELLAPANDI P reports a relationship with Bharathidasan University that includes: employment. CHELLAPANDI P has patent pending to NIL. Cou-authors are research scholars of corresponding author].

Data availability

Data will be made available on request.

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Appendix A. Supplementary material

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.gene.2023.147402>.

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